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1 Introduction

GROningen Mixture of Alchemy and Childrens' Stories

from coolstuff.cpp

1.1 Introduction

I highly recommend to follow one sentence per line.

Contrary to popular belief, Lorem Ipsum is not simply random text. It has roots in a piece of classical Latin literature from 45 BC, making it over 2000 years old. Richard McClintock, a Latin professor at Hampden-Sydney College in Virginia, looked up one of the more obscure Latin words, consectetur, from a Lorem Ipsum passage, and going through the cites of the word in classical literature, discovered the undoubtable source. Lorem Ipsum comes from sections 1.10.32 and 1.10.33 of "de Finibus Bonorum et Malorum" (The Extremes of Good and Evil) by Cicero, written in 45 BC. This book is a treatise on the theory of ethics, very popular during the Renaissance. The first line of Lorem Ipsum, "Lorem ipsum dolor sit amet..", comes from a line in section 1.10.32.

Figure 1.1 : Give your caption here.

1.2 Some equations

$$m_i \frac{d^2\mathbf{r}_i}{dt^2} = \mathbf{F}_i(\mathbf{r}), \quad i = 1, \ldots, N \,. \tag{1.1}$$

Here, m_i and $\mathbf{r}_i$ are the mass and position of the ith atom, and $\mathbf{F}_i$ is the net force acting on that atom due to the presence of all other atoms. The forces are the negative derivatives of a potential function $V(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N)$:

$$\mathbf{F}_i = -\frac{\partial V}{\partial \mathbf{r}_i}. \tag{1.2}$$

1.3 Outline of the thesis

Write something here before starting the below.

Chapter 2 investigates the full-length tertiary structure and conformational dynamics of monomeric DNAJB6b. Using homology modeling and microsecond-long all-atom molecular dynamics simulations, we show that DNAJB6b transitions between three conformational states: a closed state, an open state, and a less frequent extended state. The closed and open states are the predominant conformations observed during the simulations. Across all three states, the autoinhibitory state, i.e., the interdomain interaction between helix V in the G/F domain and helices II and III of the J-domain remains intact, suggesting that the autoinhibitory mechanism persists in the absence of substrate, potentially preventing Hsp70 binding under non-stressed cellular conditions.

And do this for all chapters here.

2 Chapter 2

Your theory is crazy, but it's not crazy enough to be true.

from coolstuff.cpp

It is a long established fact that a reader will be distracted by the readable content of a page when looking at its layout. The point of using Lorem Ipsum is that it has a more-or-less normal distribution of letters, as opposed to using 'Content here, content here', making it look like readable English. Many desktop publishing packages and web page editors now use Lorem Ipsum as their default model text, and a search for 'lorem ipsum' will uncover many web sites still in their infancy. Various versions have evolved over the years, sometimes by accident, sometimes on purpose (injected humour and the like).

This chapter has been published as: V. Adupa, E. Ustyantseva, H. H. Kampinga, P. R. Onck. "*Tertiary structure and conformational dynamics of the anti-amyloidogenic chaperone DNAJB6b at atomistic resolution*", Nature Communications 15, 3285 (2024).

2.1 Introduction

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Figure 2.1 : **Figure title here (a)** It is a long established fact that a reader will be distracted by the readable content of a page when looking at its layout. The point of using Lorem Ipsum is that it has a more-or-less normal distribution of letters, as opposed to using 'Content here, content here', making it look like readable English. Many desktop publishing packages and web page editors now use Lorem Ipsum as their default model text, and a search for 'lorem ipsum' will uncover many web sites still in their infancy. Various versions have evolved over the years, sometimes by accident, sometimes on purpose (injected humour and the like). *(Caption continued on next page.)*

Figure 2.1 *caption (continued).* Contrary to popular belief, Lorem Ipsum is not simply random text. It has roots in a piece of classical Latin literature from 45 BC, making it over 2000 years old. Richard McClintock, a Latin professor at Hampden-Sydney College in Virginia, looked up one of the more obscure Latin words, consectetur, from a Lorem Ipsum passage, and going through the cites of the word in classical literature, discovered the undoubtable source. Lorem Ipsum comes from sections 1.10.32 and 1.10.33 of ”de Finibus Bonorum et Malorum” (The Extremes of Good and Evil) by Cicero, written in 45 BC. This book is a treatise on the theory of ethics, very popular during the Renaissance. The first line of Lorem Ipsum, ”Lorem ipsum dolor sit amet..”, comes from a line in section 1.10.32.

3 Chapter 3

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This chapter is based on: V. Adupa, M. K. Musskopf, T. Bergsma, E. Ustyantseva, L. M. Veenhoff, H. H. Kampinga, P. R. Onck. "*Dual defense mechanism of the molecular chaperone DNAJB6b against polyglutamine aggregation*". In preparation.

3.1 Introduction

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4 Chapter 4

I Solve Problems

from coolstuff.cpp
Pulp Fiction

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This chapter is based on: V. Adupa, P. R. Onck. "*Oligomerization of DNAJB6b is driven by the substrate binding domains while leaving J-domain autoinhibition unaffected*". In preparation.

4.1 Introduction

Contrary to popular belief, Lorem Ipsum is not simply random text. It has roots in a piece of classical Latin literature from 45 BC, making it over 2000 years old. Richard McClintock, a Latin professor at Hampden-Sydney College in Virginia, looked up one of the more obscure Latin words, consectetur, from a Lorem Ipsum passage, and going through the cites of the word in classical literature, discovered the undoubtable source. Lorem Ipsum comes from sections 1.10.32 and 1.10.33 of "de Finibus Bonorum et Malorum" (The Extremes of Good and Evil) by Cicero, written in 45 BC. This book is a treatise on the theory of ethics, very popular during the Renaissance. The first line of Lorem Ipsum, "Lorem ipsum dolor sit amet..", comes from a line in section 1.10.32.

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5 Chapter5

Talk is cheap. Show me the code.

Linus Torvalds

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This chapter is based on: V. Adupa, J. D. Polet, M. Dekker, P. R. Onck. "*Understanding the role of DNAJB6b in Huntington's disease through coarse-grained molecular dynamics*". In preparation.

5.1 Introduction

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Table 5.1 : **Conformational sampling of DNAJB6b.** The final composition of states of DNAJB6b in the simulations starting with mixed, all-closed, all-open, and all-extended DNAJB6b. The percentages of states are given for the final 50 ns of the simulations.

		Closed state [%]			Open state [%]			Extended state [%]		
DNAJB6b†	Q48†	R1	R2	R3	R1	R2	R3	R1	R2	R3
Mixed	Solution	45	46	39	15	12	4	39	42	57
	Droplet	40	39	41	12	11	8	47	50	51
	Amyloid	39	39	38	14	8	9	47	54	53
All closed	Solution	99	99	100	1	1	0	0	0	0
	Droplet	99	99	100	1	1	0	0	0	0
	Amyloid	98	100	100	2	0	0	0	0	0
All open	Solution	2	8	1	14	16	14	85	76	86
	Droplet	2	0	2	22	14	19	77	86	79
	Amyloid	1	0	2	23	15	30	77	84	68
All extended	Solution	3	3	1	12	14	19	85	82	81
	Droplet	0	1	0	14	24	10	86	75	90
	Amyloid	0	2	0	23	18	6	77	80	93

† Initial configuration of the molecules.

5.1.1 Hydrophobicities of amino acids

Table 5.2 : **Hydrophobicities of amino acids.** Three-letter codes, one-letter codes and hydrophobicities ε_i of the 20 common amino acid types taken from the 1BPA v1.1 coarse-grained model[1].

Amino acid type	Three-letter code	One-letter code	ε_i	q_i $[e]$
Alanine	ALA	A	0.7	0
Arginine	ARG	R	0.005	+1
Asparagine	ASN	N	0.41	0
Aspartatic acid	ASP	D	0.005	−1
Cysteine	CYS	C	0.68	0
Glutamine	GLN	Q	0.33	0
Glutamatic acid	GLU	E	0.005	−1
Glycine	GLY	G	0.48	0
Histidine	HIS	H	0.53	0
Isoleucine	ILE	I	0.98	0
Leucine	LEU	L	1	0
Lysine	LYS	K	0.005	+1
Methionine	MET	M	0.78	0
Phenylalanine	PHE	F	1	0
Proline	PRO	P	0.65	0
Serine	SER	S	0.45	0
Threonine	THR	T	0.51	0
Tryptophan	TRP	W	0.96	0
Tyrosine	TYR	Y	0.82	0
Valine	VAL	V	0.94	0

6 Epilogue: Kind of summary and open questions in the field

GROningen MAchine for Chemical Simulation - GROMACS

from coolstuff.cpp

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6.1 Chapter 2 title here

In **Chapter 2**, ...

6.2 Chapter 3 title here

In **Chapter 3**, ...

6.3 Chapter 4 title here

In **Chapter 4**, ...

6.4 Chapter 5 title here

In **Chapter 5**, ...

6.5 Open questions

.....

Summary

All good things must come to an end.

Summary of all the chapters should go here.

Nederlandstalige samenvatting

> ’T sal waerachtig wel gaen
>
> *Willem Barendsz*

same thing in your native language or dutch.

Acknowledgements

Most important part of the thesis and the one you should write all the way at the end.

List of publications

As part of this thesis

Chapter 2: V. Adupa, E. Ustyantseva, H. H. Kampinga, P. R. Onck. "Tertiary structure and conformational dynamics of the anti-amyloidogenic chaperone DNAJB6b at atomistic resolution." In: Nature Communications 15.1 (2024), p. 3285.

Outside of this work

- if you got anything here, put that here.

Bibliography

[1] M. Dekker, E. van der Giessen, and P. R. Onck. "Phase Separation of Intrinsically Disordered FG-Nups Is Driven by Highly Dynamic FG Motifs". In: *Proceedings of the National Academy of Sciences of the United States of America* 120.25 (2023), p. 9. DOI: 10.1073/pnas.2221804120 (cit. on p. 24).